REACT REDUX REFRESH

Day 1 to Day 11 Interview Guide

A printable study guide built from your full progress so far, from Day 1 through Day 11, March 23, 2026. It covers React fundamentals, TypeScript, routing, local state, performance, Tailwind, feature architecture, Redux Toolkit, async thunks, normalized state, selectors, Redux DevTools, testing strategy, and production frontend architecture patterns.

Coverage: Day 1, Day 2, Day 3, Day 4, Day 5, Day 6, Day 7, Day 8, Day 9, Day 10, and Day 11 now cover RTK Query, fetchBaseQuery, optimistic mutations, rollback safety, URL-driven server-state screens, paginated caching, and runtime schema validation for safer frontend contracts. Update this file after each new learning day and regenerate the PDF.

1. What You Should Be Able To Explain

Core React

- How props flow down and events flow up.
- How state updates trigger re-renders.
- Why controlled inputs are predictable.
- How reconciliation uses element type and keys.
- When memoization helps and when it is noise.

State Architecture

- What belongs in local component state.
- What belongs in shared global state.
- Why selectors are preferable to ad hoc derivation everywhere.
- Why normalized state scales better than large nested arrays.
- Why async state needs loading and error modeling.
- How Redux Toolkit reduces accidental complexity.
- How to test reducers, thunks, and Redux-connected UI separately.
- How route boundaries, lazy loading, and profiling support production frontend scale.
- How RTK Query replaces repetitive async thunk boilerplate for server state.
- How optimistic updates, rollback, DTO mapping, and fetchBaseQuery make server-state code more production-shaped.
- How URL query state, pagination, and prefetching change scalable list-screen architecture.
- How schema validation protects forms, query params, and API payloads at runtime.

2. Roadmap Through Today

Day	Main focus	What changed in the project
Day 1	React foundations	Built the task app with routing, local state, controlled forms, and basic localStorage persistence.

Day	Main focus	What changed in the project
Day 2	Performance and reuse	Added reusable components, Tailwind styling, and practical memoization with `React.memo`, `useMemo`, and `useCallback`.
Day 3	Redux Toolkit basics	Moved shared task state into Redux Toolkit with slices, typed hooks, and selector-driven reads.
Day 4	Async state	Introduced `createAsyncThunk`, loading and error modeling, and a storage service boundary for async flows.
Day 5	Advanced Redux patterns	Normalized task state with entity adapter, split selectors and thunks into dedicated modules, improved async boundaries, and configured Redux DevTools tracing.
Day 6	Testing Redux and UI	Added Vitest and React Testing Library, wrote reducer and thunk tests, mocked the storage API boundary, tested form behavior, and verified Redux + UI flows together.
Day 7	Production frontend architecture	Refactored routing around feature-level route objects, added lazy route and panel loading with Suspense, introduced a shared app shell, deferred non-critical task views, and added dev-only profiling boundaries.
Day 8	RTK Query for server state	Replaced manual thunk-based fetching and mutations with an RTK Query API slice, generated hooks, cache invalidation, and endpoint-focused tests.
Day 9	Production-ready RTK Query flows	Moved to fetchBaseQuery, added shared API config, DTO mapping, normalized API errors, optimistic create/toggle/delete updates, rollback verification, and stronger loading, retry, and empty states.
Day 10	Scalable query-driven list screens	Moved task search, filter, and page state into the URL, upgraded getTasks into a paginated RTK Query endpoint, added toolbar and pagination components, prefetched adjacent pages, and refined query-specific loading and empty states.
Day 11	Runtime-safe validation boundaries	Added zod schemas for task contracts, validated query params and form input, parsed request and response DTOs at runtime, normalized schema failures, and extended tests for invalid query, form, and transport cases.

3. Day 1 Foundations

Component Model

A React component is a function that maps state and props to UI. It is not a template file plus code-behind. Every render is a fresh execution of the component function. Variables declared inside the component are recalculated on each render unless persisted through hooks or lifted out of the render path.

Props, Events, and Data Flow

```
Parent state
  -> passes props to child
Child reads props
  -> user interacts
Child calls callback prop
  -> parent updates state
Parent re-renders
  -> updated props flow down again
```

This is the default React model: unidirectional data flow. The parent owns the state, children receive values and functions. Senior-level reasoning here is about keeping ownership explicit and not scattering mutable state across unrelated components.

Controlled Inputs

A controlled input binds `value` to React state and updates that state in `onChange`. The source of truth becomes the component state instead of the DOM node. This makes validation, formatting, submit handling, and reset logic deterministic.

> **Interview question:** Why use controlled inputs instead of uncontrolled inputs?
>
> **Answer:** Controlled inputs make state transitions explicit, which improves validation, serialization, reset behavior, and coordination with other UI state. Uncontrolled inputs can be simpler for basic forms, but controlled inputs are the standard choice when the form behavior matters.

Local Storage Persistence

Local storage is synchronous and browser-local. It works for demos, simple offline persistence, and learning projects, but it is not a substitute for a real server. It has no conflict resolution, auth, schema evolution, or concurrent-user guarantees.

React Router Basics

Router configuration maps URLs to UI. At interview level, explain that route state should be used for navigation concerns, not as a generic replacement for application state. Route params identify resources; app state represents current client data.

4. React Internals You Should Know

Concept	What it means	Why interviewers care
Render phase	React computes the next UI tree by calling components.	Shows you understand render is calculation, not DOM mutation.
Commit phase	React applies the minimal DOM updates after diffing.	Helps explain why pure rendering matters.
Reconciliation	React compares previous and next trees.	Explains keys, preservation of state, and remount behavior.
Batching	Multiple state updates can be grouped into one render.	Shows you know React is scheduling work, not immediately mutating UI.
Closures	Handlers capture values from the render they were created in.	Important for stale state bugs and hook dependencies.

> **Interview question:** Why does React need keys in lists?
>
> **Answer:** Keys let React track identity across renders. Without stable keys, React may reuse the wrong DOM nodes or component instances, causing incorrect state preservation and unnecessary remounting.

> **Interview question:** What causes a React component to re-render?
>
> **Answer:** A re-render happens when its local state changes, its parent re-renders and passes new props, the component consumes changed context, or a subscribed store selector returns a changed value.

5. How Params and State Move Through a React App

Props

Use props when state ownership is local to a parent and children only need values or callbacks. This is the cleanest path because the dependency graph stays obvious.

```
type TaskFormProps = {
  onAdd: (title: string) => void;
  disabled?: boolean;
};
```

Local State

Use `useState` for UI concerns that are local and temporary: input text, modal open/closed state, touched flags, active tab, or draft values that do not need broad sharing.

Route Params

Use route params to identify the resource the screen should display: `/tasks/:taskId`. Do not put large mutable UI state in the URL unless shareability and navigation semantics justify it.

Redux Store

Use Redux when multiple parts of the app depend on the same state, when update logic should be centralized, or when async request status needs a consistent model. Components should read store state through selectors, not by repeatedly digging through the store shape inline.

```
UI event
  -> dispatch(thunk or action)
Store processes reducer / extraReducer
  -> new state produced
Selector reads derived result
  -> subscribed component re-renders
```

How To Choose

Mechanism	Use when	Avoid when
Props	State has a clear nearby owner.	Too many intermediate components are forwarding data.
Local state	The data is view-local and short-lived.	Other screens or features need it.
Route params	The URL should identify current resource or filter.	The data is ephemeral UI detail.
Redux	Multiple consumers, shared async status, centralized rules.	The state is purely local and simple.

6. Day 2 Performance and Reuse

`React.memo`

`React.memo` skips re-rendering a child when its props are shallowly equal. It is useful when the child is expensive or receives stable props. It is not a default wrapper for every component.

`useMemo`

Use `useMemo` for expensive derived values or to maintain stable references when reference identity matters. Do not use it to hide cheap calculations or as cargo-cult optimization.

`useCallback`

`useCallback` memoizes function identity. It matters when a child is memoized or when a dependency array would otherwise change on every render. It does not make the function faster to execute.

> **Interview question:** When is `useCallback` useless?
>
> **Answer:** It is usually useless when the callback is cheap, the child is not memoized, and function identity is not part of any dependency-sensitive path. In those cases, it adds code without reducing work.

Feature-Based Architecture

A feature folder groups components, state, utilities, and types by business domain rather than by technical layer. This improves local reasoning, reduces cross-folder hunting, and makes ownership clearer.

Tailwind CSS

Tailwind is utility-first styling. Senior-level usage means being able to discuss tradeoffs: fast iteration and predictable colocation versus large class strings and the need for disciplined abstractions when repeated patterns emerge.

7. TypeScript in a React Codebase

- Type props at the component boundary.
- Type event handlers where DOM event detail matters.
- Type Redux hooks so selectors and dispatch stay safe.
- Use unions for finite UI states such as filters and request status.
- Prefer explicit domain types over `any`.

> **Interview question:** Why is `any` dangerous in frontend code?
>
> **Answer:** It disables the compiler exactly at the boundary where bugs are likely: props, API data, reducer payloads, and event handling. It lowers trust in refactors and makes runtime issues easier to ship.

8. Day 3 Redux Toolkit

Store, Slice, Reducer, Selector

Redux Toolkit provides structured primitives for predictable state management. A store holds application state. A slice defines a domain of state plus reducers. Selectors are read APIs over store state. This separation matters because it

prevents UI components from becoming the place where every rule lives.

Immer and Immutable Updates

Redux Toolkit uses Immer under the hood, allowing reducer code that looks mutative while producing immutable results. The practical benefit is less boilerplate and lower risk of accidental mutation bugs.

```
state.items.unshift(newTask)

// looks mutable, but Immer tracks the draft and
// returns a safe immutable next state
```

Why Selectors Matter

Selectors centralize read logic. If every component calculates visible tasks, stats, or derived flags separately, business logic becomes duplicated and inconsistent. Selectors also support memoization.

> **Interview question:** What problem does Redux Toolkit solve over plain Redux?
>
> **Answer:** It removes repetitive setup, encourages sane defaults, bakes in Immer, simplifies async with thunks, and improves maintainability. It solves accidental ceremony more than conceptual complexity.

9. Day 4 Async Thunks and Request State

Why Async Changes the Design

Once data is asynchronous, the UI can no longer assume a value is simply present. You must model at least four states: idle, loading, succeeded, and failed. Without this, your app will show stale screens, broken controls, or silent failures.

`createAsyncThunk` Lifecycle

```
dispatch(fetchTasks())
  -> pending
  -> async work runs
  -> fulfilled with payload
     or rejected with error
```

A thunk is not just "an async function in Redux." It is a structured flow that gives reducers lifecycle events, lets UI track request state, and keeps side effects outside presentational components.

Senior-Level Concerns in Async UI

- Prevent overlapping writes when the UI cannot safely merge them.
- Show clear loading and error states.
- Keep reducer logic deterministic.
- Model retries intentionally, not accidentally.
- Know when local optimistic updates are safe and when they are not.

> **Interview question:** Why store request status in Redux state?
>
> **Answer:** Because loading, success, and failure are part of the user-visible state of the feature. Multiple components may need to react consistently to those states, so centralizing them avoids fragmented UI behavior.

> **Interview question:** What is the difference between a reducer and a thunk?
>
> **Answer:** A reducer is a pure state transition function. A thunk orchestrates side effects and dispatches actions before and after async work. Reducers must stay deterministic; thunks are where I/O belongs.

10. Day 5 Advanced Redux Patterns

Normalized State with Entity Adapter

Storing entities as an ID map plus ordered IDs is a production pattern because reads and writes stay predictable as the collection grows. Instead of replacing an entire array after each mutation, reducers can add, update, or remove a single entity with focused operations.

```
tasks: {
  ids: ["t3", "t2", "t1"],
  entities: {
    t1: { id: "t1", title: "Ship Redux refactor", completed: false },
    t2: { id: "t2", title: "Write selectors", completed: true },
    t3: { id: "t3", title: "Inspect DevTools", completed: false }
  }
}
```

Selectors as the Read API

Components should not know storage shape details. A selector layer becomes the read API for the feature: visible task IDs, counts, loading flags, and derived error state. Memoization matters when selectors are reused and when their outputs would otherwise allocate new arrays or objects every render.

> **Interview question:** Why memoize selectors instead of computing filtered lists inside components?
>
> **Answer:** Because selector memoization centralizes derivation, avoids repeated allocation work for identical inputs, and keeps components focused on rendering instead of reimplementing domain rules.

Cleaner Separation of Concerns

A maintainable Redux feature usually separates write orchestration, state transitions, and read logic:

- `tasksApi.ts` now handles server-state reads, writes, and cache invalidation.
- `tasksSlice.ts` handles state transitions.
- `tasksSelectors.ts` handles derived read logic.
- `storage.ts` acts as the persistence boundary.

Better Async Patterns

Production thinking means modeling fetch and mutation status separately, preventing overlapping writes when the UI cannot safely merge them, and preserving useful failure messages with `rejectWithValue`. This makes the UI and DevTools timeline easier to reason about.

Redux DevTools as an Architecture Tool

DevTools is not just for debugging bugs after the fact. It helps verify that actions are small, reducers are deterministic, thunk lifecycles are visible, and state shape stays understandable. If your DevTools history looks noisy or confusing, the architecture often needs work.

What Today's Refactor Proves

- You can treat selectors as the stable read API for a feature.
- You can update one entity without replacing the entire collection.
- You can separate fetch status from mutation status instead of collapsing everything into one loading flag.
- You can keep side effects in thunks and state transitions in reducers.
- You can use DevTools to inspect architecture quality, not just bugs.

> **Interview question:** What is the benefit of entity adapter if the UI still shows a simple list?
>
> **Answer:** The UI may be simple today, but normalized state reduces update cost, improves selector composition, and avoids rewriting reducers later when detail views, lookups, or cross-feature references appear.

> **Interview question:** Why separate thunk files from slice files?
>
> **Answer:** Because async orchestration and pure state transitions are different responsibilities. Separating them reduces file sprawl, makes testing easier, and keeps reducers focused on deterministic updates.

11. Day 6 Testing Redux and UI

What Was Added in the Task App

Day 6 turned the architecture work into a testing surface. The project now has four complementary test layers:

- `tasksSlice.test.ts` verifies pure reducer transitions such as filter changes and query hydration.
- `tasksApi.test.ts` verifies endpoint behavior, cache invalidation, and RTK Query query and mutation flows.
- `TaskForm.test.tsx` verifies component behavior such as validation, trimmed input submission, and field reset behavior.
- `TasksPage.integration.test.tsx` verifies the full Redux-connected screen: initial loading, fetched tasks, adding a task, toggling a task, and last-mutation UI updates.

Why the Storage Layer Is Mocked

The task feature uses `storage.ts` as a persistence boundary. In tests, that boundary is mocked with `vi.mock` so the test can control responses without depending on real browser persistence or timer delays. That keeps tests focused on Redux and UI behavior rather than localStorage mechanics.

```
UI event
  -> generated mutation hook or dispatched action runs
```

```
Mocked storage boundary responds
  -> query cache or slice state updates
Selector output changes
  -> component re-renders with updated UI
```

How To Explain the Testing Strategy

The testing strategy is layered. Pure reducer tests prove deterministic state transitions. Thunk tests prove async orchestration and error translation. Component tests prove isolated UI behavior. Integration tests prove that Redux state, selectors, thunks, and rendering work together under user interaction.

> **Interview question:** Why unit test reducers separately if integration tests already exist?
>
> **Answer:** Reducers are pure and cheap to test, so they give fast feedback on state transition rules. Integration tests are broader and valuable, but reducer tests make it easier to pinpoint whether a failure is in state logic or in the UI wiring.

> **Interview question:** What should an async thunk test prove?
>
> **Answer:** It should prove the orchestration contract: whether the thunk calls the dependency with the right inputs, whether condition guards block invalid work, and whether fulfilled or rejected results produce the expected Redux state and payloads.

> **Interview question:** Why mock the API or storage boundary instead of using the real implementation in every test?
>
> **Answer:** Because most tests should validate application behavior, not re-test browser storage mechanics or network timing. Mocking the boundary makes tests deterministic, faster, and more focused on the code under review.

> **Interview question:** What is the difference between a component test and a Redux + UI integration test?
>
> **Answer:** A component test isolates a small UI unit and usually passes props directly. A Redux + UI integration test renders the connected screen with a real store, dispatch flow, selectors, and user events so it can validate that the feature works as a system.

> **Interview question:** In this task app, what does the Day 6 integration test actually prove?
>
> **Answer:** It proves that the page loads task data, renders loading and fetched results, submits a new task through the connected Redux and RTK Query flow, toggles an existing task, and reflects the last mutation in the UI. That is stronger evidence than checking one reducer or one component in isolation.

> **Interview question:** How do you keep frontend tests from becoming brittle?
>
> **Answer:** Test behavior instead of implementation details. Prefer user-visible queries, keep mocks at clear boundaries, avoid asserting incidental markup, and reserve deep internal assertions for pure units like reducers or selector logic.

12. Day 7 Production Frontend Architecture

What Changed in the Task App

Day 7 moved the project from a small learning app toward a more production-shaped frontend architecture. The changes were focused on delivery boundaries, route ownership, and measurable rendering cost.

- `app/routes.tsx` now composes a real router with `createBrowserRouter`.
- `features/tasks/routes.tsx` owns the task feature route definition.
- `App.tsx` became a shell with `Outlet` instead of manually embedding pages.
- `PageLoader.tsx` and `LoadingPanel.tsx` provide reusable loading boundaries.
- `TasksPage.tsx` defers non-critical list updates and lazy-loads the insights panel.
- `RenderProfiler.tsx` adds dev-only `React.Profiler` measurements without polluting tests.

Why Code Splitting Matters

Code splitting is not just a build trick. It is an architectural decision about what the browser must pay for up front and what can wait until a route or secondary UI is actually needed. In this project, the task page and the insights panel now load as separate chunks instead of being folded into the first app bundle.

Lazy Loading and Suspense

`React.lazy` delays module loading until the component is requested, and `Suspense` gives the UI a controlled fallback while that code arrives. Senior-level reasoning here is about choosing loading boundaries intentionally so the user sees continuity instead of jarring blank screens.

```
navigate to route
  -> router matches feature route
  -> lazy component chunk requested
  -> Suspense fallback renders
  -> chunk resolves
  -> feature screen mounts
```

Feature-Level Routing

Feature-level routing means the feature exports its own route objects while the app composes them into the top-level router. That keeps route ownership closer to the feature and scales better when more domains are added later.

Performance Profiling and Deferred UI

Production architecture should be observable, not just theoretically optimized. The project now uses a dev-only profiler wrapper and `useDeferredValue` for the task list so expensive or secondary rendering paths can be inspected and softened before they become user complaints.

> **Interview question:** What is the difference between code splitting and lazy loading?
>
> **Answer:** Code splitting is the build-time outcome of breaking code into separate chunks. Lazy loading is the runtime strategy that decides when a chunk should be requested. They are related, but not the same layer of the system.

> **Interview question:** Why use `Suspense` around a route boundary?
>
> **Answer:** Because route transitions are natural loading boundaries. Wrapping a route in `Suspense` lets the app show a stable fallback while the lazy route chunk loads, which is cleaner than letting the screen flash empty or hand-rolling loading logic in every page.

> **Interview question:** What is the benefit of feature-level routing?
>
> **Answer:** It keeps route ownership close to the feature, reduces top-level router sprawl, and makes it easier to add or remove feature screens without forcing the whole app router file to become a dumping ground.

> **Interview question:** Why defer a value like visible task IDs?
>
> **Answer:** Because the user interaction can stay responsive while React prepares a lower-priority visual update. It is useful when a derived list may be large or when you want filter changes to feel smoother without changing the source of truth.

> **Interview question:** Why add profiling hooks in development?
>
> **Answer:** Profiling turns performance conversations from guesswork into measurement. It helps you see whether memoization, deferred rendering, or route boundaries are actually reducing work instead of just adding complexity.

> **Interview question:** What makes component architecture "advanced" in this project?
>
> **Answer:** The app now separates shell layout, route boundaries, loading primitives, feature orchestration, and secondary insight panels into distinct layers. That separation is what makes scaling easier, not just the number of files.

13. Day 8 RTK Query for Server State

Why RTK Query Exists

Manual thunk handling works, but it tends to repeat the same patterns: request lifecycle flags, duplicate loading state, error translation, cache refresh logic, and hand-written orchestration between reads and writes. RTK Query exists to standardize those server-state concerns so the feature code can focus more on domain behavior and less on plumbing.

What Changed in the Task App

- `tasksApi.ts` now owns the feature API slice.

- `getTasks` is modeled as a query.
- `addTask`, `toggleTask`, `deleteTask`, and `clearCompleted` are modeled as mutations.
- Cache tags are used so writes invalidate the task list automatically.
- `TasksPage.tsx` now reads and writes through generated hooks.
- `tasksApi.test.ts` verifies query fetching, mutation execution, and invalidation behavior directly.

Queries vs Mutations

A query reads data and is cacheable. A mutation changes data and usually invalidates one or more cached queries so fresh data is fetched again. That distinction is one of the main conceptual upgrades over a generic thunk-based async function.

```
useGetTasksQuery()
  -> cached task list
useAddTaskMutation()
  -> write to storage layer
  -> invalidate Task LIST tag
  -> getTasks refetches automatically
```

Caching and Invalidation

RTK Query uses cache tags to understand which reads are affected by a write. In this project, task mutations invalidate the task list tag so the feature does not need manual refetch dispatches after every add, toggle, delete, or clear-completed action.

Loading and Error States

The task page now gets fetch loading and fetch errors from the RTK Query hooks, while keeping only the local mutation metadata that still serves the UI. This is a cleaner separation between server-state lifecycle and feature-specific view metadata.

> **Interview question:** What is RTK Query?
>
> **Answer:** RTK Query is the data fetching and caching layer built into Redux Toolkit. It standardizes queries, mutations, caching, invalidation, loading state, and generated hooks for server-state workflows.

> **Interview question:** Why use RTK Query instead of manual thunks?
>
> **Answer:** Because it removes repetitive async boilerplate, centralizes cache behavior, gives you generated hooks, and handles refetching through invalidation rules instead of forcing every feature to reinvent those patterns manually.

> **Interview question:** What is the difference between a query and a mutation in RTK Query?
>
> **Answer:** A query reads and caches data. A mutation performs a write and often invalidates cached queries so related data can be refreshed automatically.

> **Interview question:** What problem do cache tags solve?
>
> **Answer:** They let the data layer know which cached results are stale after a mutation. That means the app can refetch the right queries automatically instead of dispatching manual refresh logic after every write.

> **Interview question:** What is an auto-generated hook in RTK Query?
>
> **Answer:** It is a React hook generated from an endpoint definition, such as `useGetTasksQuery` or `useAddTaskMutation`. Those hooks expose data, loading flags, errors, and trigger functions without hand-written wrapper code.

> **Interview question:** How does RTK Query differ from thunks conceptually?
>
> **Answer:** Thunks are a general async orchestration tool. RTK Query is a specialized server-state layer with opinions about caching, invalidation, subscriptions, and generated hooks. Thunks can still be useful, but RTK Query is usually a better fit for standard CRUD-style data fetching.

14. Day 9 Production-Ready RTK Query Flows

What Changed in the Task App

Day 9 pushed the RTK Query setup closer to production frontend architecture. The feature now uses `fetchBaseQuery`-style requests, a shared API config module, DTO mapping between transport and domain models, normalized error payloads, optimistic cache updates, rollback coverage, and stronger retry and empty-state UX.

- `apiConfig.ts` now centralizes API base URL and task resource configuration.
- `taskDtos.ts` separates API payload shapes from the UI task model.
- `tasksHttp.ts` acts as a fetch-compatible transport adapter around the storage boundary.
- `tasksApi.ts` now uses `fetchBaseQuery`, DTO transforms, and optimistic cache patches.
- `apiErrors.ts` normalizes query and mutation failures into UI-safe messages.
- `tasksApi.test.ts` now verifies rollback behavior for failed optimistic mutations.

Why DTO Mapping Matters

A backend payload is not automatically the right shape for the UI. DTO mapping keeps transport concerns such as field naming or date serialization separate from the domain model the app actually renders. That makes backend changes easier to absorb without leaking transport details through the feature.

Why Use `fetchBaseQuery`

`fetchBaseQuery` gives RTK Query a more realistic request model with HTTP-style methods, URLs, status-aware failures, and easier migration to a real backend later. In this project, the fetch transport still routes through the storage boundary, but the feature code now looks like a real API-driven frontend.

Optimistic Updates and Rollback

Day 9 introduced optimistic create, toggle, and delete mutations. The cache is patched immediately so the interface feels fast, and each mutation keeps an undo handle so the optimistic state can roll back if the request fails. This is a better

production pattern than waiting for every request before updating the UI.

```
mutation starts
  -> patch cached task list immediately
  -> UI updates right away
request succeeds
  -> keep or reconcile optimistic data
request fails
  -> undo cache patch
  -> show normalized error message
```

Loading, Retry, and Empty States

Production UX is not just about successful requests. The task page now distinguishes between initial blocking load failures, background refreshes, filter-specific empty states, and retry actions. That makes the feature more resilient and easier to explain in an interview.

> **Interview question:** Why introduce DTOs if the app only has one task model?
>
> **Answer:** Because transport models and UI models change for different reasons. DTOs let the API evolve without forcing the whole UI and selector layer to adopt backend naming or serialization choices directly.

> **Interview question:** What is the benefit of `fetchBaseQuery` over a custom fake query?
>
> **Answer:** It moves the feature toward real HTTP semantics, gives you standard request configuration, and makes it easier to replace a mock transport with a real backend later without rewriting the endpoint layer.

> **Interview question:** When should you patch cache data instead of invalidating and refetching?
>
> **Answer:** Patch the cache when you want immediate UX feedback and you can safely predict the local result, such as add, toggle, or delete. Invalidate and refetch when the server is the source of truth for more complex recalculation or when local prediction is risky.

> **Interview question:** Why is rollback important for optimistic updates?
>
> **Answer:** Because optimistic UI without rollback can leave the client in a false state when the request fails. Rollback preserves responsiveness without sacrificing correctness.

> **Interview question:** What should a production-ready loading and error experience include?
>
> **Answer:** It should distinguish first-load blocking failures from background refreshes, provide retry paths, keep existing data visible when safe, and make empty states specific enough to explain what the user can do next.

15. Day 10 URL-Driven Paginated Task Screens

What Changed in the Task App

Day 10 turned the task screen into a more production-shaped list page. Query state now lives in the URL, the RTK Query endpoint accepts paginated arguments, transport and domain mapping now cover paginated responses, and the UI includes reusable toolbar and pagination controls with adjacent-page prefetching.

- `TasksPage.tsx` now reads filter, search, and page from `useSearchParams`.
- `tasksApi.ts` now models `getTasks` as a paginated query keyed by `TaskListQuery`.
- `taskDtos.ts` and `tasksHttp.ts` now map and transport paginated task payloads.
- `TasksQueryToolbar.tsx` and `TasksPagination.tsx` split search/filter and paging UI into reusable boundaries.
- `TasksPage.integration.test.tsx` now verifies URL-driven search, page navigation, and paginated optimistic mutations together.

Why URL Query State Matters

Search, filter, and page are navigation concerns as much as UI state. Putting them in the URL makes the screen shareable, refresh-safe, and easier to reason about when the user moves between pages or sends a link to someone else. This is a common production requirement for list screens.

Paginated RTK Query and Cache Keys

Once a query depends on page, search, and filter, the cache key is no longer just `getTasks`. Each query-arg combination represents a separate cached result. That is why the task endpoint now accepts a full `TaskListQuery` object and why optimistic updates need to know which page view they are patching.

```
URL search params
  -> TaskListQuery
  -> useGetTasksQuery(query)
  -> cache key includes page + search + filter
page changes
  -> new query arg
  -> separate cached result or fetch
prefetch next page
  -> warmer navigation path
```

Prefetching and Stable UX

The task page now prefetches adjacent pages when metadata says they exist. That does not guarantee every navigation is instant, but it improves perceived performance for common next-page and previous-page moves. Combined with query-aware empty and refresh states, the list feels much closer to a real data screen.

> **Interview question:** Why store search and pagination state in the URL instead of local component state?
>
> **Answer:** Because they affect navigation and shareability. URL state survives refreshes, supports deep links, and more accurately models a list page than hidden local state does.

> **Interview question:** How do RTK Query query arguments affect caching?
>
> **Answer:** Each unique argument combination creates a distinct cache entry. If page, search, or filter changes, RTK Query treats that as a different query result and caches it separately.

> **Interview question:** Why prefetch the next page of a paginated list?
>
> **Answer:** It reduces the wait the user feels during common navigation paths. Prefetching is useful when the next action is predictable and the additional network cost is acceptable.

> **Interview question:** What changes when optimistic updates are used on paginated data?
>
> **Answer:** The mutation needs to know which cached page or query context to patch. Optimistic logic becomes view-aware because the user may be looking at only one filtered page of a larger dataset.

> **Interview question:** What makes a list screen architecture scalable?
>
> **Answer:** Clear separation between URL state, query args, transport mapping, cache behavior, and UI controls. That keeps search, filter, pagination, and loading logic from collapsing into one oversized component.

16. Day 11 Runtime-Safe Frontend Contracts

What Changed in the Task App

Day 11 hardened the task feature with runtime validation. Instead of trusting raw query strings, form input, persisted task data, and API payloads, the app now parses those values through shared `zod` schemas before they shape UI state.

- `taskSchemas.ts` now defines the runtime contracts for tasks, queries, DTOs, and mutation payloads.
- `TaskForm.tsx` now validates titles through schema parsing instead of only checking for empty strings.
- `TasksPage.tsx` now sanitizes invalid query params into safe defaults before fetching.
- `taskDtos.ts`, `tasksHttp.ts`, and `storage.ts` now parse request, response, and persisted data through schemas.
- `apiErrors.ts` now normalizes schema-validation failures into UI-safe messages.

Why Runtime Validation Matters

TypeScript only protects the code the compiler can see. It does not validate browser storage, URL params, or network payloads at runtime. That gap matters because frontend apps constantly receive untrusted data from outside their own component tree.

```
raw user or transport input
  -> schema parse
  -> valid typed data enters state flow
invalid input
  -> normalized validation message
  -> feature stays in a safe state
```

What Improved Architecturally

The app now has clearer trust boundaries. Query parsing, DTO conversion, and form submission each validate at their own edge, which reduces hidden assumptions and makes failures easier to explain in both code review and interviews.

Interview question: Why is TypeScript alone not enough for API safety?

Answer: Because TypeScript disappears at runtime. The compiler cannot stop malformed JSON, invalid query params, or corrupted local storage from reaching the app, so runtime parsing is still necessary at trust boundaries.

Interview question: Where should schema validation happen in a frontend app?

Answer: At the boundaries where unknown data enters the system: URL params, form submissions, storage reads, and API request or response transforms. That keeps invalid data from leaking deeper into state and UI logic.

Interview question: Why validate both form input and API payloads if they represent the same task title?

Answer: Because they are different trust boundaries. The form protects the user interaction layer, while the API contract protects the transport layer in case another caller bypasses the form or sends malformed data.

Interview question: What is the benefit of centralizing schema failure normalization?

Answer: It gives the UI one predictable error language instead of leaking raw parser details everywhere. That improves maintainability and keeps validation failures understandable to users and developers.

Interview question: What changed about the task query flow after validation was added?

Answer: The URL query still drives the list, but invalid page, filter, and search values are now sanitized before the fetch happens. That means the list page stays shareable without trusting every URL blindly.

17. Senior Frontend Interview Questions and Answers

Q: How do you decide whether state should live in a component, context, or Redux?

A: Start with ownership and scope. If one component owns it, keep it local. If a tree branch needs it and prop drilling becomes structural noise, context may fit. If multiple parts of the app need the data plus coordinated updates, async lifecycle, selectors, or debugging support, Redux is justified.

Q: What would make you reject a memoization optimization in code review?

A: If it is guarding cheap work, added without measurement, obscures code, or relies on unstable dependencies that invalidate it every render. Memoization should remove meaningful work, not add cognitive overhead for theoretical wins.

Q: How would you explain stale closures to a junior developer?

A: A callback remembers variables from the render where it was created. If you later expect it to see newer values without updating its dependencies or using the right pattern, it can act on old data.

Q: Why are selectors better than reading nested store paths directly in every component?

A: They centralize read logic, improve reuse, isolate store shape changes, and support memoization. Components should ask for domain data, not know too much about storage layout.

Q: How would you describe Immer in one sentence?

A: Immer lets you write reducer logic in a mutating style while safely producing immutable next state behind the scenes.

Q: What are the risks of using index as a React list key?

A: If list order changes, component identity shifts incorrectly, which can preserve the wrong local state or cause unnecessary DOM churn. Index keys are only acceptable for static, never-reordered lists.

Q: How would you discuss localStorage in a system design interview?

A: As a browser persistence primitive suitable for lightweight local caching or user convenience, but not as a source of truth for multi-user, secure, or highly consistent systems.

Q: What should a good loading state do?

A: Communicate that work is in progress, prevent broken interactions, preserve context where possible, and avoid visual jumps that confuse the user.

Q: How do you evaluate whether Redux is overkill?

A: If the shared state is tiny, short-lived, or consumed by a narrow subtree, Redux can add indirection. If the feature requires centralized updates, async modeling, selector reuse, or debugging clarity, it is often justified.

Q: What is the difference between derived state and duplicated state?

A: Derived state is computed from a source of truth when needed. Duplicated state stores the same information in multiple places, which creates synchronization bugs.

18. How To Explain Your Project in an Interview

A strong answer is structured. Do not just list libraries. Explain the problem, architectural choices, and tradeoffs.

```
I built a task management learning project in React and TypeScript over eleven structured days.
I started with local state and controlled forms to reinforce data flow.
Then I added Tailwind and reusable components while practicing memoization.
After that I moved shared task state into Redux Toolkit with typed hooks and selectors.
For Day 4, I introduced createAsyncThunk and modeled request status, loading, and error handling using
an async local storage layer.
For Day 5, I normalized the task state with entity adapter, split selectors and thunks into dedicated
modules, and configured Redux DevTools tracing so the state transitions are easier to inspect.
For Day 6, I added Vitest and React Testing Library, then wrote reducer tests, async-flow tests,
component tests, and a Redux + UI integration test for the task page.
For Day 7, I refactored the app around feature-level routing, lazy-loaded route and secondary panel
chunks, introduced a shared app shell, and added profiling-aware render boundaries to make performance
decisions more observable.
For Day 8, I replaced manual thunk-based fetching and mutations with RTK Query, using an API slice,
generated hooks, cache invalidation, and endpoint-focused tests to simplify server-state handling.
For Day 9, I moved that setup toward production-style API architecture with fetchBaseQuery, DTO
mapping, normalized API errors, optimistic cache updates, rollback coverage, and stronger retry and
empty-state UX.
For Day 10, I turned the task page into a URL-driven paginated list screen with query-param state,
paginated RTK Query caching, reusable toolbar and pagination controls, adjacent-page prefetching, and
more specific loading and empty states.
For Day 11, I added runtime-safe validation with zod so task forms, query params, DTO mapping,
persisted data, and RTK Query transport boundaries all parse untrusted input before it affects state.
The main goal was not feature count; it was learning how state architecture, testing, delivery
boundaries, and server-state tooling evolve together as complexity grows.
```

19. Red Flags Interviewers Notice

- Using `any` at critical boundaries.
- Memoizing everything by default.
- Confusing route state with application state.
- Putting business logic in random components.
- No loading or error states for async operations.
- Using unstable keys in dynamic lists.
- Duplicating derived state in multiple places.
- Replacing whole entity collections for small mutations.
- Choosing Redux before proving shared-state need.
- Not being able to explain why a hook dependency exists.
- Writing brittle tests that assert implementation details instead of behavior.
- Using integration tests only and skipping focused unit tests for pure logic.
- Talking only about syntax instead of tradeoffs.
- Adding lazy loading everywhere without choosing sensible loading boundaries.
- Keeping manual thunk boilerplate after RTK Query already owns the server-state workflow.
- Using optimistic updates without a rollback path when the request can fail.
- Keeping search, filter, and page state hidden in local state when the screen should be linkable.
- Ignoring query-key design when paginated or filtered server data is cached.
- Trusting raw query params, local storage, or API payloads without runtime validation.

20. Final Revision Checklist

- Can you explain props, state, and controlled inputs clearly?
- Can you describe render, commit, reconciliation, and keys?
- Can you justify when to use `useMemo` or `useCallback`?
- Can you explain why feature-based architecture scales better?

- Can you describe store, slice, reducer, selector, and Immer?
- Can you explain `createAsyncThunk` pending, fulfilled, and rejected?
- Can you explain normalized state, IDs/entities, and entity adapter tradeoffs?
- Can you explain why selectors form the read API for Redux state?
- Can you explain how Redux DevTools helps validate architecture decisions?
- Can you explain what reducers, thunks, component tests, and integration tests each prove?
- Can you explain why mocking the API boundary improves determinism?
- Can you explain how code splitting, lazy loading, and Suspense work together?
- Can you explain why feature-level routing scales better than one giant router file?
- Can you explain when deferred rendering helps and when it is unnecessary?
- Can you explain what RTK Query is and why it exists?
- Can you explain the difference between a query and a mutation?
- Can you explain how caching and invalidation work in RTK Query?
- Can you explain how auto-generated hooks reduce boilerplate?
- Can you explain when DTO mapping protects the UI from backend shape changes?
- Can you explain when optimistic cache patching is better than invalidation alone?
- Can you explain why rollback is necessary for optimistic mutations?
- Can you explain why URL query params are a better fit than local state for searchable paginated lists?
- Can you explain how RTK Query query args change cache keys and refetch behavior?
- Can you explain when adjacent-page prefetching improves UX and when it is unnecessary?
- Can you explain how parameters and state move through your app?
- Can you explain why schema validation is still necessary in a TypeScript app?
- Can you explain where runtime parsing belongs in a frontend architecture?
- Can you discuss tradeoffs like a senior engineer rather than just list APIs?

Suggested next update after Day 11: replace the fetch adapter with a real backend, extend schema validation beyond tasks, add richer server-backed list metadata, and add end-to-end browser coverage for validation-heavy flows.